

BRODY DICKSON

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EDUCATION

Northeastern University Boston, MA

Candidate for Bachelor of Science in Mechanical Engineering; Expected May 2029

Relevant Coursework: Mechanics of Materials, Physics I-II, Thermodynamics, Acoustics and Psychoacoustics of Sound

Activities: American Society of Mechanical Engineers, AerospaceNU, DJing, Golf, Intramural Dodgeball and Kickball

TECHNICAL SKILLS

Design & Analysis Software: SolidWorks, AutoCAD, MATLAB, Microsoft Excel, Ghost Arcade (Visual Projection Software)

Programming: Python, Swift, SwiftUI, HTML, MicroPython, Claude, Claude Skills, ChatGPT

Fabrication & Machining: Lathe, Milling, Laser Cutting, 3D Printing, Welding, Soldering, Woodworking, Small Electronics

PROJECTS

Breathalyzer Lock Box *Northeastern University Cornerstone, Team Project, Jan-May 2026*

- Designed and built a pocket-sized key lockbox that releases car keys only after a fingerprint match and an estimated BAC under 0.06, aimed at reducing drunk driving in rural college towns
- Modeled the enclosure and servo-driven locking mechanism in SolidWorks; programmed the MQ-3 alcohol sensor, fingerprint scanner, and OLED display in MicroPython on a Raspberry Pi Pico
- Analyzed sensor output data to improve the calibration and accuracy of a low-cost MQ-3 sensor, and built lockout logic and driver logging to block bypass attempts.

Vintage Motorcycle Rebuild *Independent Team Project, Feb-April 2026*

- Built a custom motorcycle by integrating an imported crate engine into a vintage steel bike frame, designing the build around parts never made to fit together.
- Fabricated engine, gas tank and gear mounts using metal working tools to fit the engine and drivetrain to the frame

ML-Optimized Autonomous RC Car *Independent Project, December-May 2025*

- Engineered an autonomous RC vehicle applying OpenCV to real-time navigation, path planning, and obstacle avoidance without manual input.

WORK EXPERIENCE

Dock Hand 2026 - Present (Summer and Fall), *Boston Electric Boats, Boston, MA*

- Maintain and troubleshoot mechanical issues on operating boats to ensure safe, reliable performance.
- Deliver hands-on customer service, resolving client issues on the water to sustain satisfaction and team morale.

Founder & Owner 2022 - 2025 (All Seasons), *BD Landscaping Greater Chicago Area, IL*

- Founded and scaled a full-service landscaping business to 120+ clients through door-to-door outreach and targeted Facebook and Nextdoor marketing
- Built a data-driven pricing model by tracking labor hours by job phase and property condition, enabling accurate on-the-spot quotes and a rate increase that kept most existing clients
- Hired, trained, and managed 9 employees, adding a second foreman so two crews could run jobs at the same time
- Identified offseason opportunities and expanded into snow removal, moving, and interior work for off-season revenue; owned all invoicing, payroll, overhead, and budgeting

Computer & Small Electronics Repair Technician 2023 - 2025 (Winters), *Gadget Bar, Barrington, IL*

- Diagnosed and repaired hardware and software issues across phones, laptops, and small electronics.
- Developed custom repair solutions tailored to individual client needs, driving repeat business.

Honor Caddy 2020 - 2023, *Biltmore Country Club, Barrington, IL*

- Earned Honor Caddy distinction through consistent professionalism, course knowledge, and elite client service

LEADERSHIP & VOLUNTEER EXPERIENCE

FFA Member & Community Farm Coordinator 2021 - 2024

- Co-founded and built a community farm, overseeing infrastructure development, crop planning, and day-to-day agricultural operations, and educated community members on sustainable farming practices and agricultural techniques.
- Managed produce sales operations from harvest to point-of-sale, developing business and logistics skills.